CTCF binding architecture of clustered TF-paralog loci: ohnolog vs. small-scale-duplication (tandem) clusters

Iteration 19 — Paralogous TF project
ETS P01 genomics platform

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Abstract

We ask whether CTCF (insulator / TAD-boundary) binding differs between the two *clustered* classes of the iteration-16 four-group TF-duplication catalog: **clustered cis-ohnolog** loci (arising by whole-genome duplication) versus **clustered SSD-tandem** loci (tandem small-scale duplications). Using a single ENCODE GM12878 CTCF ChIP-seq conservative-IDR peak set (hg38), we profiled CTCF over each clustered locus extended by ±100 kb. Both groups are CTCF-enriched relative to the genome-wide expectation (∼1.5–1.9× obs/exp), and global CTCF density is statistically indistinguishable between them (Mann–Whitney U, all $p > 0.7$). The one metric that separates the two groups is the *number of CTCF peaks lying between the two paralog members* (median 3 for ohnolog vs. 1 for SSD-tandem; $p = 0.0035$), but this difference is largely explained by the larger inter-member spacing of ohnolog pairs: once normalized to inter-member distance, the per-kb CTCF density between members is no longer significant ($p = 0.27$). The interpretation is that ohnolog paralog pairs are separated by *more* insulator elements because they sit farther apart, not because the intervening DNA is intrinsically denser in CTCF.

1 Data and provenance

Duplication catalog (reused, not recomputed). The four-group catalog was taken verbatim from iteration 16 (`TF_dup_2x2_classification.tsv`) together with the clustered-locus coordinates (`clustered_paralogs.noKRABHOX.tsv`); genome build hg38 (GENCODE v45 basic). Every cluster in the clustered file maps to a single, consistent `dup_class4` label (0 mixed clusters). This iteration analyses only the two *clustered* classes; dispersed classes are out of scope.

CTCF peaks (one download). ENCODE experiment `ENCSR000DZN` (TF ChIP-seq, target CTCF, biosample GM12878, EBV-transformed B-lymphoblastoid), file `ENCFF796WRU` — *conservative IDR-thresholded peaks*, `bed narrowPeak`, assembly GRCh38, released. Downloaded 2026-07-18 from encodeproject.org. Sanity checks: 44,217 peaks; chromosomes chr1–22 and chrX (no chrY, as expected for the female GM12878 line); point-ish narrowPeak intervals. Accession and date are recorded in `log/encode_download.log` and in the script headers. GM12878 was chosen for its lymphoid relevance to the project; the cell line is a CLI argument so a second line can be swapped in later.

2 Methods

Windows. Each clustered locus was extended to [`cluster_start` − 100 kb, `cluster_end` + 100 kb], clipped to hg38 chromosome bounds (1 window shortened by clipping). One chrY locus (HSFY1/HSFY2, SSD-tandem) was dropped because GM12878 carries no chrY signal and would contribute a technical zero. Final set: **47 clustered cis-ohnolog** and **30 clustered SSD-tandem** clusters (counts are *clusters*, not member TFs; caveat ii).

Binning. Two representations were produced (caveat iii). (A) Fixed-width 5 kb bins across the raw window (used only for the per-locus heatmap; bins do not align across loci of different span). (B) A length-normalized *metagene*: the left 100 kb flank in 20 fixed-bp bins, the cluster body in 20 scaled bins, the right 100 kb flank in 20 fixed-bp bins, on a relative-position axis $-1 \to 0$ (left flank), $0 \to 1$ (body), $1 \to 2$ (right flank). The metagene is what is averaged and compared across loci.

CTCF metrics. Per bin: peak-summit count and bp-coverage fraction. Per locus: CTCF peaks/kb over the window, body, and flanks; observed/expected ratio against the genome-wide baseline (44,217/3.03 Gb = 0.0146 peaks/kb); distance of the nearest CTCF peak to each member TSS; and the count of CTCF peaks between the outermost member TSSs (both raw and normalized per kb of inter-member distance).

Statistics (reproducibility rule). Two-sided **Mann–Whitney U**, **clustered cis-ohnolog** vs. **clustered SSD-tandem**, on each per-locus continuous metric (report U, p, per-group n and medians, rank-biserial effect size r_{rb}). Metagene profiles carry bootstrap 95% CI bands (2000 resamples). A per-bin Mann–Whitney U across the 60 metagene bins was run with Benjamini–Hochberg FDR to localize any positional difference.

3 Results

Both clustered classes are CTCF-enriched, and globally indistinguishable

Over the ±100 kb window both groups carry CTCF well above the genome-wide expectation (median obs/exp 1.70 ohnolog, 1.52 SSD-tandem). None of the global density metrics separate the groups (Table 1; all $p > 0.70$), and the metagene profiles overlap within their bootstrap CI bands across the whole window (Fig. 1); **0 of 60** metagene bins reach BH-FDR < 0.05. In short, at the level of overall insulator density the two kinds of clustered loci occupy statistically similar CTCF environments.

Table 1: Per-locus CTCF metrics, **clustered cis-ohnolog** ($n = 47$) vs. **clustered SSD-tandem** ($n = 30$). Two-sided Mann–Whitney U. r_{rb} = rank-biserial effect size (negative ⇒ ohnolog > SSD-tandem).

Metric	median (ohnolog)	median (SSD)	U	p	r_{rb}
CTCF peaks/kb, window	0.0248	0.0222	695.0	0.921	+0.014
CTCF peaks/kb, locus body	0.0238	0.0332	668.0	0.703	+0.053
CTCF peaks/kb, flanks	0.0250	0.0225	695.0	0.921	+0.014
Observed/expected, window	1.702	1.521	695.0	0.921	+0.014
Nearest CTCF to a TSS (bp, min)	4241	1885	769.5	0.504	−0.092
Nearest CTCF to TSS (bp, mean)	11909	13727	687.0	0.855	+0.026
CTCF peaks between members	**3.0**	**1.0**	**981.5**	**0.0035**	**−0.392**
Inter-member TSS distance (bp)	74355	49409	866.0	0.094	−0.228
CTCF peaks/kb between members (norm.)	0.0232	0.0178	809.5	0.274	−0.148

Ohnolog members are separated by more CTCF peaks — a spacing effect

The single metric that distinguishes the groups is the number of CTCF peaks *between* the two paralog members: ohnolog pairs have a median of 3 intervening CTCF peaks versus 1 for SSD-tandem pairs (Mann–Whitney U, $p = 0.0035$, $r_{rb} = -0.39$; Fig. 2, left). However, ohnolog members also sit farther apart (median inter-member TSS distance 74 kb vs. 49 kb, $p = 0.09$). When the between-member CTCF count is normalized to inter-member distance, the difference is no longer significant (0.023 vs. 0.018 peaks/kb, $p = 0.27$; Fig. 2, right). The extra insulator load between ohnolog paralogs is therefore best read as a consequence of their wider genomic separation rather than an intrinsically CTCF-denser intervening region — consistent with ohnolog pairs, retained from whole-genome duplication, having drifted farther apart than recent tandem duplicates while remaining within the same broadly insulated neighbourhood.

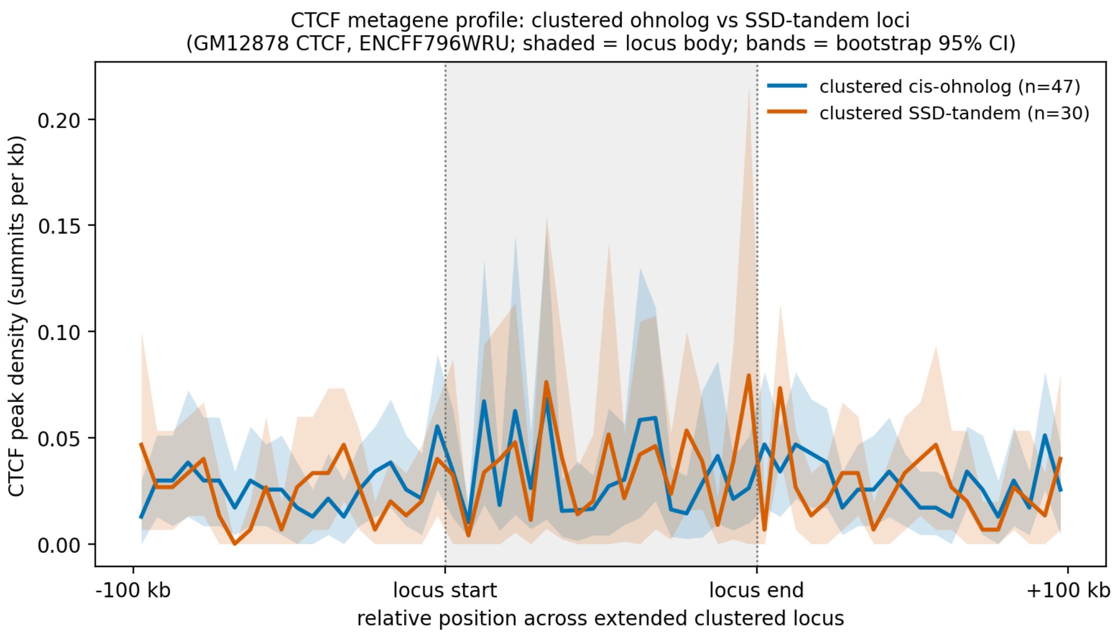

Figure 1: Length-normalized CTCF metagene profile. Lines = mean CTCF summit density per kb; shaded bands = bootstrap 95% CI; grey = locus body. The two clustered classes overlap throughout; no metagene bin survives BH-FDR < 0.05.

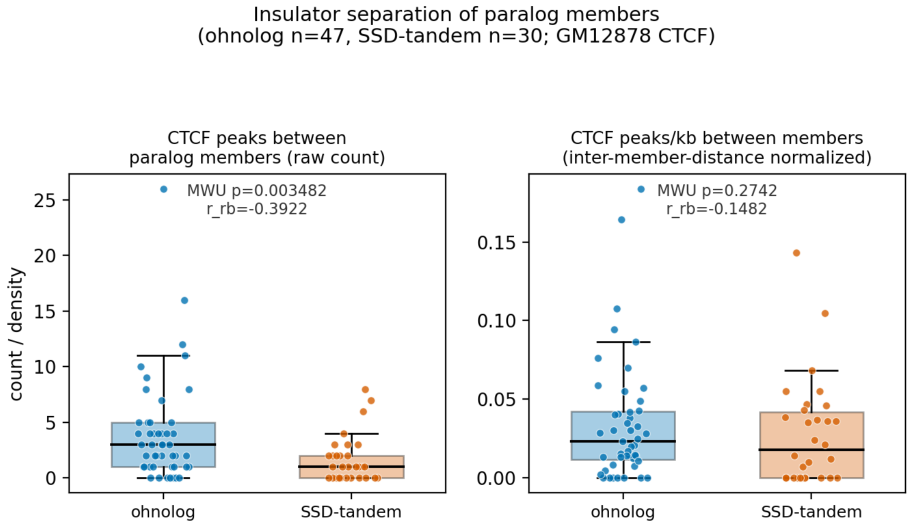

Figure 2: CTCF insulator separation of paralog members. Left: raw count of CTCF peaks between the two members differs (MWU $p = 0.0035$). Right: after normalizing to inter-member distance the per-kb difference is not significant ($p = 0.27$), indicating the raw effect is driven by the wider spacing of ohnolog pairs.

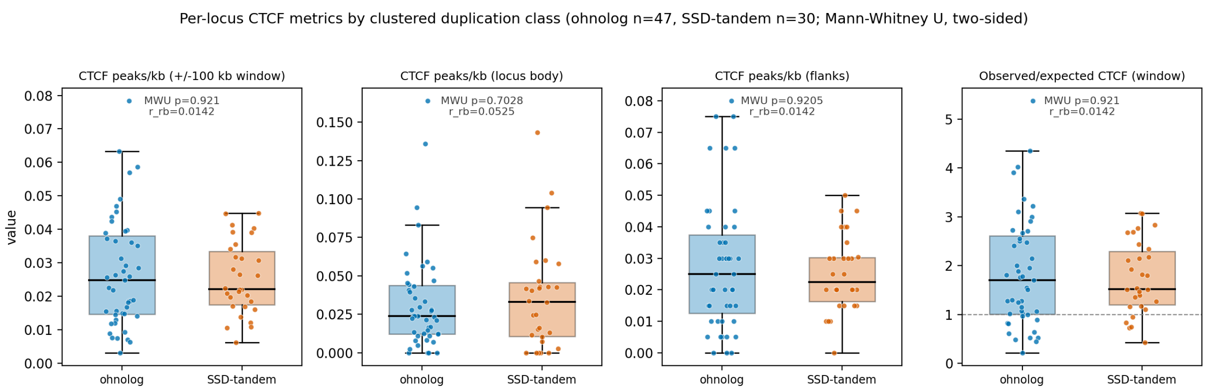

Figure 3: Per-locus CTCF density metrics by class (box + jittered points). All four global metrics are indistinguishable between ohnolog and SSD-tandem loci; both groups exceed the genome-wide expectation (dashed line = obs/exp = 1).

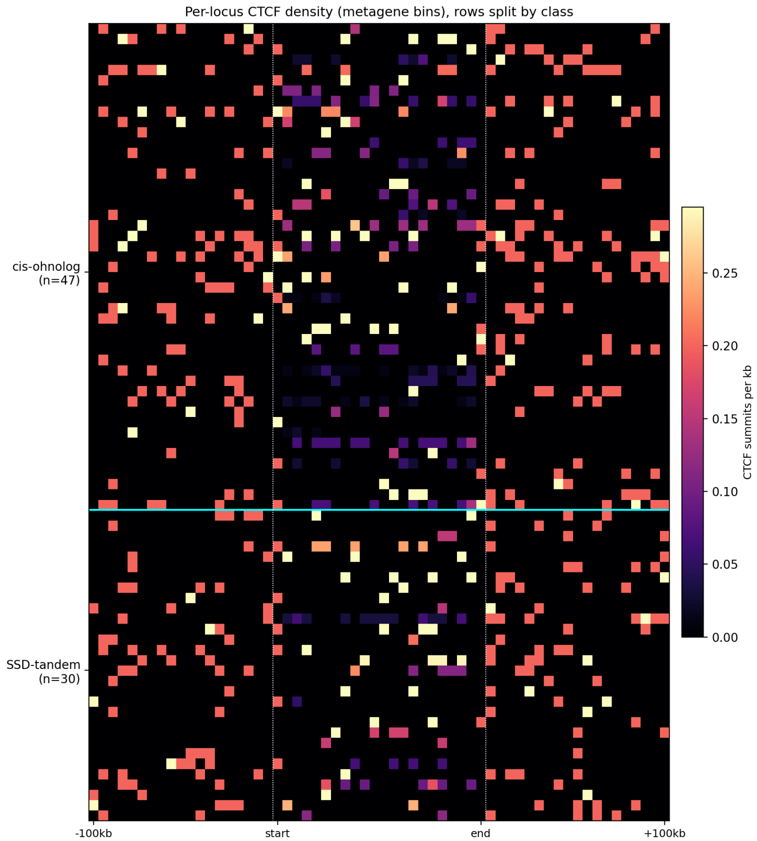

Figure 4: Per-locus CTCF density across metagene bins, rows split by class (cyan line). CTCF signal concentrates in and around the locus body in both groups without an obvious class-specific pattern.

4 Conclusion

In GM12878, clustered ohnolog and clustered SSD-tandem TF-paralog loci occupy *statistically similar* CTCF insulator environments: both are CTCF-enriched over the genome-wide background and their overall peak densities, flank/body distributions, and TSS-proximal CTCF are indistinguishable. The only separating signal — more CTCF peaks between ohnolog members — is accounted for by the greater physical distance between ohnolog paralogs and does not survive distance normalization. Within the power available (47 vs. 30 clusters) we find no evidence that the two clustered duplication mechanisms sit in distinct CTCF/insulator architectural niches.

Caveats. (i) A single cell type (GM12878); CTCF is largely constitutive but not identical across lineages — results are conditioned on this line. (ii) Small groups (47 vs. 30 clusters after collapsing members to `cluster_id`); per-locus tests are underpowered, and the borderline inter-member-distance trend ($p = 0.09$) may reflect this. (iii) Fixed-bp binning and length-normalized metagene binning are reported separately, as they are not comparable across loci of different span. (iv) Extended windows can overlap for nearby clusters and can run off chromosome ends; one window was clipped, and the single chrY locus was excluded (no chrY signal in the female GM12878 line).